1. The Data Link Layer is divided by IEEE 802 LAN standards into which two distinct functional sublayers?

(A) Logical Link Control (LLC) and Media Access Control (MAC)
(B) Physical Sublayer and Routing Sublayer
(C) Transport Sublayer and Network Sublayer
(D) Session Sublayer and Presentation Sublayer

2. The process of encapsulating a network layer packet into a discrete transmission unit with headers and trailers is:

(A) Framing
(B) Segmentation
(C) Modulation
(D) Paging

3. Which framing technique inserts special predetermined flag byte sequences (e.g., 01111110) to mark frame boundaries?

(A) Byte-Oriented Framing with Byte (Character) Stuffing
(B) Bit-pattern multiplexing
(C) Clock drift locking
(D) Manchester pulse padding

4. In Bit-Oriented Framing (HDLC), bit stuffing prevents flag mimicry (01111110) by inserting a 0 after every sequence of:

(A) Five consecutive 1s
(B) Six consecutive 1s
(C) Four consecutive 1s
(D) Seven consecutive 1s

5. A 48-bit physical hardware address permanently burned into a Network Interface Card (NIC) ROM is the:

(A) Media Access Control (MAC) Address (Ethernet Address)
(B) IP Address
(C) Port Number
(D) URL

6. A standard 48-bit MAC address is conventionally represented in human-readable format as:

(A) 12 hexadecimal digits separated by colons or hyphens (e.g., 00:1A:2B:3C:4D:5E)
(B) 4 decimal octets separated by dots
(C) 32 binary bits
(D) 8 ASCII characters

7. The first 24 bits (3 bytes) of an Ethernet MAC address represent the:

(A) Organizationally Unique Identifier (OUI) assigned to the hardware manufacturer
(B) Device serial number
(C) Network subnet identifier
(D) Broadcast domain flag

8. The special broadcast MAC address used to send an Ethernet frame to ALL nodes on the local physical LAN segment is:

(A) FF:FF:FF:FF:FF:FF
(B) 00:00:00:00:00:00
(C) 255.255.255.255
(D) AA:BB:CC:DD:EE:FF

9. The simplest error detection scheme that appends a single binary bit to ensure the total count of 1s is even or odd is:

(A) Parity Check (Even/Odd Parity)
(B) Cyclic Redundancy Check (CRC)
(C) Checksum
(D) Hamming Code

10. A simple parity check bit can detect:

(A) Any odd number of single-bit transmission errors, but completely fails on double-bit errors
(B) All two-bit burst errors
(C) All error patterns without exception
(D) Errors and automatically correct them

11. Which powerful mathematical error detection polynomial code is universally implemented in Ethernet and Wi-Fi frame trailers?

(A) Cyclic Redundancy Check (CRC-32)
(B) One's complement checksum
(C) Parity check matrix
(D) Viterbi convolutional code

12. In CRC error detection, binary division of the data polynomial by a generator polynomial G(x) is performed using:

(A) Modulo-2 Arithmetic (XOR operations without carry/borrow)
(B) Standard decimal arithmetic
(C) Floating-point division
(D) Binary 2's complement subtraction

13. In an (n, k) block error-correcting code, the minimum Hamming distance d_min required to DETECT up to 't' bit errors is:

(A) d_min >= t + 1
(B) d_min >= 2 * t + 1
(C) d_min >= t
(D) d_min >= 2 * t

14. In an (n, k) block code, the minimum Hamming distance d_min required to forward-CORRECT up to 't' bit errors is:

(A) d_min >= 2 * t + 1
(B) d_min >= t + 1
(C) d_min >= t
(D) d_min >= 3 * t

15. Hamming 7,4 Code is a linear error-correcting block code that encodes 4 data bits into 7 code bits, capable of:

(A) Single Error Correction and Double Error Detection (SEC-DED)
(B) Correcting 3 simultaneous errors
(C) Quad bit correction
(D) Zero error detection

16. Flow Control at the Data Link Layer is essential to prevent:

(A) A fast sender from overwhelming and overflowing the memory buffer of a slow receiver
(B) Packet collisions on full-duplex lines
(C) Routing loops
(D) DNS spoofing

17. In the Stop-and-Wait flow control protocol, the sender transmits one frame and then:

(A) Must halt and wait for an acknowledgment (ACK) from the receiver before transmitting the next frame
(B) Immediately sends 10 more frames
(C) Closes the connection
(D) Transmits continuously

18. In Sliding Window flow control protocols, the sender is permitted to transmit up to 'W' frames without waiting for an ACK, where W is:

(A) The Send Window Size
(B) The Maximum Transmission Unit (MTU)
(C) The Bandwidth-Delay Product
(D) The Frame Check Sequence

19. In the Go-Back-N Automatic Repeat reQuest (ARQ) protocol, if a frame is corrupted or lost, the sender must:

(A) Retransmit the lost frame AND all subsequent frames that were sent in the current window
(B) Retransmit only the single damaged frame
(C) Discard the entire transmission file
(D) Wait for receiver reset

20. In the Selective Repeat ARQ protocol, the receiver buffers out-of-order error-free frames, and the sender retransmits:

(A) ONLY the specific individual frame that was damaged or timed out
(B) All frames in the window
(C) The entire file
(D) Every alternate frame

21. The maximum window size W_s in Go-Back-N protocol with m-bit sequence numbers (modulo 2^m) must satisfy:

(A) W_s <= 2^m - 1
(B) W_s <= 2^(m - 1)
(C) W_s = 2^m
(D) W_s = m

22. In Selective Repeat ARQ with m-bit sequence numbers, the maximum window size W must satisfy:

(A) W <= 2^(m - 1)
(B) W <= 2^m - 1
(C) W = 2^m
(D) W = 2 * m

23. Carrier Sense Multiple Access (CSMA) protocols operate on which common-sense physical principle before transmitting?

(A) 'Listen before Talk' (Sense channel carrier; transmit only if channel is sensed idle)
(B) Transmit at random intervals blindly
(C) Wait for token arrival
(D) Broadcast continuously

24. In 1-Persistent CSMA, when a station finds the transmission channel busy, it:

(A) Continues sensing continuously and transmits immediately with probability p = 1 the instant the channel becomes idle
(B) Waits a fixed random backoff time
(C) Aborts transmission permanently
(D) Switches to another frequency

25. In Non-Persistent CSMA, when a station senses the channel busy, it:

(A) Waits a random period of time before sensing the channel again, reducing collision probability
(B) Senses continuously without pause
(C) Transmits anyway
(D) Sends a jam signal

26. In p-Persistent CSMA (used in slotted channels), when an idle channel is detected, the station transmits with probability 'p' and:

(A) Defers transmission to the next time slot with probability (1 - p)
(B) Aborts transmission
(C) Transmits twice
(D) Doubles power

27. CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is the classical MAC protocol implemented in wired:

(A) Half-Duplex Ethernet (IEEE 802.3)
(B) Wi-Fi (IEEE 802.11)
(C) Bluetooth (IEEE 802.15.1)
(D) Token Ring (IEEE 802.5)

28. In CSMA/CD, when two colliding stations detect simultaneous transmission, they immediately abort and broadcast a:

(A) 32-bit Jam Signal to ensure all other stations detect the collision
(B) FIN packet
(C) Reset token
(D) SYN packet

29. After collision in CSMA/CD, stations calculate their random backoff waiting delay using which algorithm?

(A) Truncated Binary Exponential Backoff Algorithm
(B) Dijkstra's shortest path algorithm
(C) Bellman-Ford algorithm
(D) Round Robin scheduling

30. In CSMA/CD, why does wireless Wi-Fi (IEEE 802.11) use CSMA/CA (Collision Avoidance) rather than CSMA/CD?

(A) Wireless transceivers cannot detect collisions while transmitting because their own transmitted RF power completely drowns out weak received signals
(B) Collision detection is illegal in wireless bands
(C) Wireless signals never collide
(D) Wi-Fi operates only in simplex mode

31. The standard minimum frame length in classic 10 Mbps and 100 Mbps Ethernet is:

(A) 64 Bytes (512 bits)
(B) 128 Bytes
(C) 1500 Bytes
(D) 32 Bytes

32. The minimum frame size requirement of 64 bytes in Ethernet CSMA/CD is mathematically required to ensure that:

(A) Transmission duration T_tx exceeds twice the round-trip propagation delay (T_tx >= 2 * T_prop) so collisions are detected before transmission completes
(B) The frame fits in the switch buffer
(C) CRC calculation is accurate
(D) Preamble synchronization is maintained

33. The standard Maximum Transmission Unit (MTU) payload size for standard Ethernet frames is:

(A) 1500 Bytes
(B) 64 Bytes
(C) 4096 Bytes
(D) 9000 Bytes

34. An Ethernet frame with payload size exceeding the standard 1500 bytes (typically up to 9000 bytes) used in enterprise data centers is a:

(A) Jumbo Frame
(B) Superframe
(C) Runt frame
(D) Baby giant frame

35. An Ethernet frame that is shorter than the minimum legal length of 64 bytes (usually resulting from an aborted collision) is called a:

(A) Runt Frame
(B) Jumbo frame
(C) Giant frame
(D) Ghost packet

36. A Layer 2 Network Switch segments a local network into multiple independent collision domains by maintaining a:

(A) MAC Address Table (Filtering Database / CAM Table)
(B) Routing Table
(C) DNS Cache
(D) ARP Cache

37. When a switch receives a frame destined for a MAC address that does not yet exist in its MAC address table, it performs:

(A) Flooding (broadcasts the frame out of all ports except the arrival port)
(B) Dropping the frame silently
(C) Sending an ICMP error
(D) Restarting the switch

38. Spanning Tree Protocol (STP, IEEE 802.1D) is essential in redundant bridged switched networks to prevent:

(A) Broadcast Storms and Layer 2 Switching Loops
(B) IP address exhaustion
(C) Buffer underruns
(D) TCP timeout stalls

39. Virtual Local Area Networks (VLANs, IEEE 802.1Q) partition a single physical switched infrastructure into multiple isolated:

(A) Logical Broadcast Domains
(B) Collision domains only
(C) Subnets sharing one MAC table
(D) Autonomous systems

40. The IEEE 802.1Q standard inserts a 4-byte VLAN tag into standard Ethernet frames, containing a 12-bit VLAN ID (VID) supporting up to:

(A) 4094 distinct VLANs (2^12 - 2)
(B) 256 VLANs
(C) 1024 VLANs
(D) 65,536 VLANs

41. In IEEE 802.3 Ethernet framing, what is the purpose of the 7-byte Preamble followed by the 1-byte Start Frame Delimiter (SFD: 10101011)?

(A) To synchronize receiver clock phase and signal the start of the MAC destination address
(B) To hold source IP address
(C) To provide encryption key
(D) To specify QoS priority

42. What is the slot time in standard 10 Mbps Ethernet defined as?

(A) 51.2 microseconds (the time to transmit 512 bits / 64 bytes)
(B) 10.0 microseconds
(C) 1.0 millisecond
(D) 100.0 nanoseconds

43. In Binary Exponential Backoff, after the k-th collision (for k <= 10), the station picks a random integer r from the range:

(A) 0 <= r <= 2^k - 1, and waits r * Slot_Time
(B) 0 <= r <= k
(C) 0 <= r <= 2 * k
(D) r = 2^k exactly

44. In CSMA/CD, after how many consecutive collisions does a station permanently abort transmission and declare an error?

(A) 16 collisions
(B) 10 collisions
(C) 8 collisions
(D) 32 collisions

45. A 1-km coaxial Ethernet cable has propagation speed v = 2 * 10^8 m/s. The one-way propagation delay T_prop is:

(A) 5 microseconds (1000 m / 2*10^8 m/s)
(B) 2 microseconds
(C) 10 microseconds
(D) 0.5 microseconds

46. For the cable in Q45, the round-trip propagation time (2 * T_prop) is 10 microseconds. At 10 Mbps, how many bits are transmitted in 10 microseconds?

(A) 100 bits (10 * 10^6 bps * 10 * 10^-6 s)
(B) 50 bits
(C) 512 bits
(D) 64 bits

47. Why is collision detection physically impossible in full-duplex switched Ethernet?

(A) Full-duplex provides dedicated, independent transmit (Tx) and receive (Rx) wire pairs, so collisions never occur
(B) Full duplex uses token passing
(C) Carrier sense is disabled permanently
(D) Switches buffer all packets

48. The 'Hidden Terminal Problem' in wireless networks (IEEE 802.11) occurs when:

(A) Two stations out of radio range of each other transmit simultaneously to a common access point, causing collision
(B) A rogue station eavesdrops on frames
(C) Antennas are misaligned
(D) Frequency bands overlap

49. CSMA/CA resolves the hidden terminal problem in wireless LANs through which optional handshake mechanism?

(A) RTS/CTS (Request to Send / Clear to Send) exchange
(B) Collision detection jamming
(C) Token ring rotation
(D) Slotted ALOHA synchronization

50. In CSMA/CA, the Network Allocation Vector (NAV) is a timer maintained by stations that acts as a:

(A) Virtual carrier-sensing mechanism indicating how long the medium will remain busy
(B) Physical signal meter
(C) Collision backoff multiplier
(D) IP routing hop counter

51. In IEEE 802.11 Wi-Fi, what is the Inter-Frame Space hierarchy from shortest to longest priority?

(A) SIFS (Shortest), PIFS (PCF), DIFS (DCF), EIFS (Extended / Error)
(B) DIFS, SIFS, PIFS, EIFS
(C) EIFS, DIFS, PIFS, SIFS
(D) PIFS, SIFS, DIFS, EIFS

52. Short Inter-Frame Space (SIFS) in 802.11 is reserved for highest-priority immediate transmissions such as:

(A) ACK frames, CTS frames, and subsequent fragments
(B) Regular data frames
(C) Beacon management frames
(D) Association requests

53. The Data Link layer error-detection checksum used in the Internet (IP, TCP, UDP headers) is calculated by:

(A) Summing 16-bit words using 1's complement arithmetic and taking the 1's complement of the result
(B) Modulo-2 CRC division
(C) XORing all bytes sequentially
(D) 2's complement negation

54. If the generator polynomial for CRC is G(x) = x^4 + x + 1, its binary divisor representation is:

(A) 10011 (coefficients of x^4, x^3, x^2, x^1, x^0)
(B) 11001
(C) 10101
(D) 11110

55. How many CRC check bits (remainder bits) are appended to a message when dividing by generator polynomial G(x) of degree r?

(A) Exactly r bits (e.g., degree 4 appends 4 zeros, produces 4 check bits)
(B) r + 1 bits
(C) r - 1 bits
(D) 2 * r bits

56. Point-to-Point Protocol (PPP, RFC 1661) is widely used for direct links between two nodes, incorporating:

(A) Link Control Protocol (LCP) for link establishment and Network Control Protocols (NCP) for network-layer multiplexing
(B) CSMA/CD collision control
(C) Token ring circulation
(D) Spanning tree routing

57. High-Level Data Link Control (HDLC) protocol defines three station types: Primary, Secondary, and Combined, and operates in which three transfer modes?

(A) Normal Response Mode (NRM), Asynchronous Response Mode (ARM), and Asynchronous Balanced Mode (ABM)
(B) Simplex, Half-Duplex, Full-Duplex
(C) Slotted, Unslotted, Hybrid
(D) Active, Standby, Idle

58. In HDLC frames, what are the three distinct frame formats defined by the control field?

(A) Information frames (I-frames), Supervisory frames (S-frames), and Unnumbered frames (U-frames)
(B) Data, Voice, Video
(C) Header, Body, Trailer
(D) Request, Response, Acknowledgment

59. In HDLC, Supervisory frames (S-frames) are used for:

(A) Flow and error control (carrying Receive Ready RR, Receive Not Ready RNR, Reject REJ)
(B) User data payload transport
(C) Session teardown
(D) Encryption keys

60. A bridge or switch learns station MAC addresses dynamically by inspecting:

(A) The SOURCE MAC address of every arriving frame and associating it with the arrival port
(B) The destination MAC address
(C) The IP header
(D) The CRC checksum

61. In Spanning Tree Protocol (STP), the root switch of the broadcast tree is elected based on the lowest:

(A) Bridge ID (BID = Priority + MAC Address)
(B) Port speed
(C) IP address
(D) VLAN tag

62. What are the five port states in legacy IEEE 802.1D Spanning Tree Protocol?

(A) Blocking, Listening, Learning, Forwarding, and Disabled
(B) Active, Passive, Up, Down, Idle
(C) Open, Closed, Half-Open, Reset, Timeout
(D) Connected, Disconnected, Error, Synced, Master

63. Rapid Spanning Tree Protocol (RSTP, IEEE 802.1w) achieves sub-second convergence primarily by introducing:

(A) Proposal-agreement handshake on point-to-point links and new port roles (Alternate and Backup)
(B) Increasing hello timers
(C) Eliminating BPDUs
(D) Using static routes

64. Link Aggregation (IEEE 802.3ad / 802.1AX - LACP) combines multiple physical Ethernet links between two switches into a single logical channel to:

(A) Increase aggregate link bandwidth and provide automatic link failover redundancy
(B) Reduce power consumption
(C) Convert copper to fiber
(D) Eliminate VLANs

65. The Address Resolution Protocol (ARP) translates an IP address into a corresponding physical MAC address using:

(A) An ARP Request broadcast (FF:FF:FF:FF:FF:FF) followed by a unicast ARP Reply from the target host
(B) A DNS server query
(C) A DHCP discover broadcast
(D) An ICMP ping

66. Gratuitous ARP is an ARP broadcast where a host announces its own IP-MAC mapping, used for:

(A) Detecting duplicate IP addresses on the LAN and updating neighboring switch/router ARP caches upon hardware replacement
(B) Requesting a DHCP lease
(C) Discovering gateway routers
(D) Testing internet connectivity

67. ARP Spoofing (ARP Poisoning) is a malicious attack wherein an attacker on the local LAN:

(A) Sends falsified ARP responses to link the attacker's MAC address with the legitimate default gateway's IP address (Man-in-the-Middle)
(B) Floods the DNS root server
(C) Disables physical network cables
(D) Cracks Wi-Fi WPA3 passwords

68. Dynamic ARP Inspection (DAI) on enterprise switches mitigates ARP spoofing by:

(A) Validating ARP packets against the DHCP snooping binding database and dropping invalid mappings
(B) Encrypting all ARP traffic with RSA
(C) Disabling ARP completely
(D) Converting all LAN traffic to IPv6

69. In an Ethernet switch, what is the architectural difference between Cut-Through switching and Store-and-Forward switching?

(A) Cut-Through forwards the frame immediately after reading the 6-byte destination MAC address; Store-and-Forward buffers the entire frame and verifies the CRC before forwarding
(B) Store-and-forward drops all frames with VLAN tags
(C) Cut-through is only supported on 10 Mbps links
(D) Store-and-forward cannot filter MAC addresses

70. Fragment-Free switching is a compromise hybrid switching mode that reads the first:

(A) 64 bytes of the frame to filter out collision fragments (runts) before forwarding
(B) 1500 bytes
(C) 12 bytes
(D) 32 bytes

71. An Ethernet Hub operates strictly at Layer 1 as a multiport repeater, meaning that:

(A) All connected devices share a single common collision domain and bandwidth is divided among all active stations
(B) Each port has its own collision domain
(C) It inspects MAC addresses
(D) It buffers frames in RAM

72. An Ethernet Switch operates at Layer 2, meaning that:

(A) Each switch port represents an isolated collision domain, enabling full-duplex dedicated wire-speed bandwidth per port
(B) All ports share one collision domain
(C) It modifies IP packets
(D) It cannot support VLANs

73. Broadcast domains in a computer network can be separated and bounded by:

(A) Routers (Layer 3) and VLANs on Layer 2 switches
(B) Layer 1 Hubs
(C) Passive patch panels
(D) Unmanaged repeaters

74. Broadcast Storms in looped switched networks cause network paralysis because Layer 2 Ethernet frames lack a:

(A) Time-to-Live (TTL) hop limit field, causing broadcast frames to circulate and multiply indefinitely
(B) Destination MAC address
(C) CRC error trailer
(D) Preamble sequence

75. IEEE 802.1X is an enterprise network port-based authentication framework that authenticates client devices using:

(A) EAP (Extensible Authentication Protocol) over LAN (EAPOL) with a central RADIUS authentication server
(B) WPA2 Pre-Shared Keys
(C) Static MAC filtering only
(D) IP address whitelisting

76. A 100 Mbps Ethernet link operates with MTU = 1500 bytes over a distance with one-way propagation delay T_prop = 1 ms. In Stop-and-Wait protocol, the link utilization efficiency is:

(A) Approximately 5.6% (T_tx = 120 us; Round trip = 2000 us; Efficiency = 120 / 2120)
(B) 50.0%
(C) 95.0%
(D) 1.2%

77. For the link in Q76 (Bandwidth = 100 Mbps, RTT = 2 ms, BDP = 200,000 bits = 16.6 frames), what is the minimum sliding window size W required to achieve 100% channel utilization?

(A) 17 frames (W >= 1 + 2 * a, where a = T_prop / T_tx = 1000 / 120 = 8.33 => 1 + 16.66 = 17.66)
(B) 8 frames
(C) 32 frames
(D) 4 frames

78. A data bit stream 1101011011 is to be transmitted using CRC with generator polynomial G(x) = x^4 + x + 1 (10011). What is the calculated 4-bit CRC remainder?

(A) 1110 (obtained by modulo-2 polynomial division of 11010110110000 by 10011)
(B) 0110
(C) 1001
(D) 0011

79. A 12-bit Hamming code (7 data bits + 4 parity bits + 1 overall parity) uses parity bits located at power-of-two bit positions (1, 2, 4, 8). If received word is 101011011010, the syndrome calculation indicates an error at bit:

(A) Bit position 6 (Syndrome = P8 P4 P2 P1 = 0110 in binary = 6 in decimal)
(B) Bit 4
(C) Bit 10
(D) No error detected

80. In a slotted ALOHA channel with 200 active terminals each generating 10 packets per minute, if slot duration is 25 ms, the total Poisson channel traffic G is:

(A) 0.833 packets per slot (200 * 10 / 60 = 33.33 pkts/s * 0.025 s/slot)
(B) 1.50 pkts/slot
(C) 0.25 pkts/slot
(D) 2.00 pkts/slot

81. For the slotted ALOHA channel in Q80 with G = 0.833, the probability of successful transmission without collision in a slot is:

(A) S = G * exp(- G) = 0.833 * exp(- 0.833) = 0.833 * 0.4347 = 0.362 (36.2%)
(B) 0.500
(C) 0.184
(D) 0.750

82. A pure ALOHA network has a maximum theoretical throughput limit of S = 1 / (2*e) = 0.184. If channel transmission rate is 56 kbps, the maximum effective delivered data throughput is:

(A) 10.30 kbps (0.184 * 56 kbps)
(B) 28.00 kbps
(C) 5.60 kbps
(D) 20.60 kbps

83. A 1 Gbps Gigabit Ethernet (1000BASE-T) link running half-duplex implements 'Carrier Extension' to extend the minimum slot time to 512 bytes (4096 bits) because:

(A) At 1 Gbps, a 64-byte frame transmits in 512 ns, which would restrict the maximum physical network cable diameter to an impractical ~10 meters
(B) To increase payload size to 9000 bytes
(C) To accommodate IPv6 headers
(D) To allow wireless repeaters

84. To avoid wasting 448 bytes of extension padding on every small frame in Gigabit Ethernet, 'Packet Bursting' allows a sender to:

(A) Transmit a burst of multiple consecutive frames without relinquishing carrier control once the first extended frame successfully contends for the channel
(B) Compress packets with LZW
(C) Switch to ATM cells
(D) Disable CRC calculation

85. In an 802.1Q tagged Ethernet frame, the 3-bit Priority Code Point (PCP) field in the Tag Control Information (TCI) defines:

(A) IEEE 802.1p Quality of Service (QoS) class of service mapping (8 priority levels from 0 to 7)
(B) The VLAN identifier
(C) The packet drop probability
(D) The routing path

86. In STP (IEEE 802.1D), Bridge Protocol Data Units (BPDUs) are transmitted periodically every:

(A) 2 seconds (Default Hello Time)
(B) 10 seconds
(C) 30 seconds
(D) 100 milliseconds

87. In standard STP, the total convergence time required for a blocked port to transition to Forwarding upon link failure is:

(A) 30 to 50 seconds (20s Max Age + 15s Listening + 15s Learning)
(B) 2 seconds
(C) 1 second
(D) 5 minutes

88. In Rapid Spanning Tree Protocol (RSTP, 802.1w), the convergence time following a topology change is reduced to:

(A) A few milliseconds to under 1 second
(B) 15 seconds
(C) 30 seconds
(D) 10 seconds

89. In an enterprise switch with a 48 Gbps non-blocking backplane switching fabric, a 24-port Gigabit Ethernet switch can achieve full wire-speed because:

(A) 24 ports * 1 Gbps * 2 (full duplex) = 48 Gbps total bidirectional throughput without internal packet drops
(B) It drops half the frames
(C) It compresses all traffic
(D) It uses half-duplex mode

90. In a token-passing ring network (IEEE 802.5) with transmission speed R = 16 Mbps and ring latency t_ring = 10 microseconds, the bit length of the ring is:

(A) 160 bits (16 * 10^6 bps * 10 * 10^-6 s)
(B) 80 bits
(C) 1600 bits
(D) 32 bits

91. In IEEE 802.11 Wi-Fi, what is the theoretical maximum data rate of single-stream 802.11n (Wi-Fi 4) using 40 MHz channel bandwidth and 64-QAM (5/6 coding)?

(A) 150 Mbps
(B) 54 Mbps
(C) 300 Mbps
(D) 433 Mbps

92. IEEE 802.11ac (Wi-Fi 5) operates exclusively in which radio frequency spectrum band?

(A) 5 GHz band (utilizing 80 MHz and 160 MHz wide channels and 256-QAM)
(B) 2.4 GHz band exclusively
(C) 60 GHz millimeter band
(D) 900 MHz ISM band

93. IEEE 802.11ax (Wi-Fi 6) improves multi-client high-density spectral efficiency by introducing cellular-grade:

(A) OFDMA (Orthogonal Frequency Division Multiple Access) with 1024-QAM and Target Wake Time (TWT)
(B) Frequency Hopping
(C) CDMA direct sequence
(D) Analog AM modulation

94. Wi-Fi 6E and Wi-Fi 7 extend operation into which pristine, uncrowded 1.2 GHz-wide radio spectrum band?

(A) 6 GHz band (5.925 GHz to 7.125 GHz)
(B) 10 GHz band
(C) 2.4 GHz band
(D) 700 MHz white space

95. In IEEE 802.11 security, WPA3 replaces the vulnerable pre-shared key four-way handshake of WPA2 with:

(A) Simultaneous Authentication of Equals (SAE, Dragonfly handshake), providing forward secrecy and offline dictionary attack immunity
(B) WEP 64-bit RC4 keys
(C) Plaintext password transmission
(D) DES encryption

96. A network switch CAM table overflow attack (MAC flooding) attempts to degrade switch security by:

(A) Flooding the switch with thousands of bogus source MAC addresses, exhausting CAM memory and forcing the switch into fail-open hub mode (broadcasting all traffic)
(B) Disabling all power supplies
(C) Overheating the CPU
(D) Short-circuiting Ethernet ports

97. Port Security on managed Ethernet switches mitigates MAC flooding attacks by:

(A) Limiting the maximum number of allowable MAC addresses learned on a port and shutting down the port upon unauthorized MAC detection
(B) Disabling full duplex
(C) Filtering IP addresses only
(D) Requiring fiber cables

98. QinQ (IEEE 802.1ad VLAN Stacking) enables telecommunication service providers to transport customer VLANs across metropolitan networks by:

(A) Adding a second outer 802.1Q Service Provider VLAN tag (S-Tag) over the inner customer VLAN tag (C-Tag)
(B) Stripping all VLAN headers
(C) Converting Ethernet to ATM
(D) Encrypting the MAC address

99. Ethernet Flow Control (IEEE 802.3x) allows a receiving station whose input buffers are nearing capacity to send a:

(A) PAUSE Frame specifying a backoff time duration during which the transmitter must halt transmission
(B) RESET frame
(C) FIN packet
(D) ICMP Quench message

100. Priority-Based Flow Control (PFC, IEEE 802.1Qbb) used in Data Center Bridging (DCB) for lossless Ethernet (RoCE / FCoE) enhances 802.3x by:

(A) Applying pause flow control independently on a per-VLAN priority class basis, preventing packet loss without head-of-line blocking for other traffic
(B) Pausing all switch ports simultaneously
(C) Dropping low priority frames unconditionally
(D) Switching to InfiniBand hardware

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the framing mechanisms used at the Data Link Layer. Detail Byte-Oriented Framing (Byte Stuffing) and Bit-Oriented Framing (Bit Stuffing in HDLC) with comprehensive illustrative examples.

Q2. Formulate Cyclic Redundancy Check (CRC) error detection. Explain modulo-2 polynomial arithmetic. A message bit stream M = 1101011011 is to be transmitted using generator polynomial G(x) = x^4 + x + 1. Calculate the transmitted frame and verify receiver error checking.

Q3. Explain the operation of CSMA/CD in wired Ethernet. Detail carrier sensing, collision detection, the 32-bit jam signal, and the Truncated Binary Exponential Backoff Algorithm.

Q4. Derive the relationship between minimum frame size, channel transmission speed, and maximum network physical cable distance in Ethernet CSMA/CD. Calculate maximum cable distance for 10 Mbps Ethernet with 64-byte minimum frame size.

Q5. Analyze Spanning Tree Protocol (STP, IEEE 802.1D). Explain why Layer 2 switching loops cause catastrophic broadcast storms, how the Root Bridge and port roles are elected, and the convergence state transitions.

Q6. Analyze Data Link Layer and Local Area Network topic #6: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q7. Analyze Data Link Layer and Local Area Network topic #7: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q8. Analyze Data Link Layer and Local Area Network topic #8: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q9. Analyze Data Link Layer and Local Area Network topic #9: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q10. Analyze Data Link Layer and Local Area Network topic #10: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q11. Analyze Data Link Layer and Local Area Network topic #11: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q12. Analyze Data Link Layer and Local Area Network topic #12: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q13. Analyze Data Link Layer and Local Area Network topic #13: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q14. Analyze Data Link Layer and Local Area Network topic #14: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q15. Analyze Data Link Layer and Local Area Network topic #15: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q16. Analyze Data Link Layer and Local Area Network topic #16: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q17. Analyze Data Link Layer and Local Area Network topic #17: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q18. Analyze Data Link Layer and Local Area Network topic #18: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q19. Analyze Data Link Layer and Local Area Network topic #19: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q20. Analyze Data Link Layer and Local Area Network topic #20: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q21. Analyze Data Link Layer and Local Area Network topic #21: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q22. Analyze Data Link Layer and Local Area Network topic #22: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q23. Analyze Data Link Layer and Local Area Network topic #23: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q24. Analyze Data Link Layer and Local Area Network topic #24: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q25. Analyze Data Link Layer and Local Area Network topic #25: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q26. Analyze Data Link Layer and Local Area Network topic #26: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q27. Analyze Data Link Layer and Local Area Network topic #27: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q28. Analyze Data Link Layer and Local Area Network topic #28: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q29. Analyze Data Link Layer and Local Area Network topic #29: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

Q30. Analyze Data Link Layer and Local Area Network topic #30: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switched network security.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

CRC treats binary data strings as polynomials with coefficients in Galois Field GF(2).
Arithmetic is performed using Modulo-2 addition and subtraction,
which is mathematically equivalent to the bitwise Exclusive-OR (XOR) logic operation without
carries or borrows:
0 + 0 = 0; 0 + 1 = 1; 1 + 0 = 1; 1 + 1 = 0.

2. Step-by-Step Transmitter Algorithm:
- Step 1: Define Generator Polynomial G(x) and its binary representation:
G(x) = x^4 + x^1 + x^0 = 1*x^4 + 0*x^3 + 0*x^2 + 1*x^1 + 1*x^0 => Binary Divisor G = 10011.
Degree of generator polynomial r = 4.
- Step 2: Append r = 4 zeros to the message bit sequence M = 1101011011:
Appended message M' = 1101011011 0000 (representing M(x) * x^r).
- Step 3: Perform Modulo-2 Polynomial Division (M' / G):
11010110110000 / 10011:
11010110110000
10011

10011
10011

000010110000
10011

010100
10011

01110 -> 4-bit Remainder R = 1110.
- Step 4: Construct Transmitted Frame T:
T = M appended with Remainder R = 1101011011 1110.

3. Receiver Verification:
The receiver receives frame T and divides it by the identical generator G = 10011 using
Modulo-2 division.
Because T(x) = M(x)*x^r + R(x), the remainder of T / G is IDENTICALLY ZERO (Remainder = 0000).
If remainder is zero, frame is accepted as completely error-free. Any non-zero remainder
indicates transmission corruption.

>>**Solution to Question 3:** Explain the operation of CSMA/CD in wired Ethernet. Detail carrier sensing, collision detection, the 32-bit jam signal, and the Truncated

stations to recognize the collision.

2. Truncated Binary Exponential Backoff Algorithm:
To prevent immediate repeated collisions upon re-sensing, each colliding station executes exponential backoff:
- Let k be the number of collisions experienced by the current frame (attempt counter).
- Set exponent variable n = min(k, 10).
- The station randomly picks an integer r uniformly from the discrete set: r in {0, 1, 2, ..., 2^n - 1}.
- The station waits backoff time T_backoff = r * Slot_Time (where Slot Time = 512 bit times = 51.2 us at 10 Mbps).
- Range Expansion with Collisions:
* After 1st collision (k=1): r in {0, 1}.
* After 2nd collision (k=2): r in {0, 1, 2, 3}.
* After 3rd collision (k=3): r in {0, 1, ..., 7}.
* After 10th collision (k=10): r in {0, 1, ..., 1023}.
* Maximum cap: For k between 10 and 15, range remains fixed at 0 to 1023.
* After 16 consecutive collisions: The station permanently aborts transmission and signals an excessive collision error to upper layers.

Solution to Question 4: >>Explain to derive the relationship between minimum frame size, channel transmission speed, and maximum network physical cable distance in Et

- In practical IEEE 802.3 standards with repeater delays, maximum network diameter is standardized at 2,500 meters (e.g., five 500m segments connected by 4 repeaters - the 5-4-3 rule).

- IEEE 802.1Q tagged frame encapsulation: TPID (0x8100) + TCI (PCP, DEI, VID).
- CSMA/CA contention window backoff mechanics.

3. Practical Enterprise Implementation:
Deployment in enterprise multi-gigabit switched backbones, virtualized software-defined networking (SDN), and wireless mesh infrastructure.

>>Solution to Question 8: Analyze Data Link Layer and Local Area Network topic #8: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switch

3. Practical Enterprise Implementation:
Deployment in enterprise multi-gigabit switched backbones, virtualized software-defined networking (SDN), and wireless mesh infrastructure.

>>Solution to Question 10: Analyze Data Link Layer and Local Area Network topic #10: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switc

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>>Solution: Question 14: Analyze Data Link Layer and Local Area Network topic #13: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switc

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>>Solution Step 1: Analyze Data Link Layer and Local Area Network topic #10: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switc

1. Protocol Foundations:
In-depth technical analysis of data link protocol standard and media access control topic #19.

2. Algorithmic and Mathematical Specifications:
- Error detection efficiency and syndrome decoding.
- IEEE 802.1Q tagged frame encapsulation: TPID (0x8100) + TCI (PCP, DEI, VID).
- CSMA/CA contention window backoff mechanics.

3. Practical Enterprise Implementation:
Deployment in enterprise multi-gigabit switched backbones, virtualized software-defined networking (SDN), and wireless mesh infrastructure.

>>Solution to Question 20: Analyze Data Link Layer and Local Area Network topic #20: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switc

2. Algorithmic and Mathematical Specifications:
- Error detection efficiency and syndrome decoding.
- IEEE 802.1Q tagged frame encapsulation: TPID (0x8100) + TCI (PCP, DEI, VID).
- CSMA/CA contention window backoff mechanics.

3. Practical Enterprise Implementation:
Deployment in enterprise multi-gigabit switched backbones, virtualized software-defined networking (SDN), and wireless mesh infrastructure.

>>Solution to Question 23: Analyze Data Link Layer and Local Area Network topic #23: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switc

- CSMA/CA contention window backoff mechanics.

3. Practical Enterprise Implementation:
Deployment in enterprise multi-gigabit switched backbones, virtualized software-defined networking (SDN), and wireless mesh infrastructure.

>>Solution to Question 28: Analyze Data Link Layer and Local Area Network topic #28: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switc

Deployment in enterprise multi-gigabit switched backbones, virtualized software-defined networking (SDN), and wireless mesh infrastructure.

>>Solution to Question 20: Analyze Data Link Layer and Local Area Network topic #20: Discuss Ethernet standards, Wi-Fi architectures, VLAN trunking, and switc